

Sachin Kalsi

Principal Data Scientist

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Seeking Staff / Principal Data Scientist roles in LLM or NLP-focused teams.

9 years in Data Science, 3 in software. Been at Draup since 2018, promoted 4 times to Principal. Lead a team of 12, mostly working on LLM and NLP systems at scale. Recent work includes building CURIE (RAG + Text-to-SQL product used by 260 enterprise clients), LLM fine-tuning pipelines on hundreds of millions of records, and setting up ML monitoring in production. Work directly with the CPO, CTO, and CEO on the AI roadmap.

Core areas:

LLMs: fine-tuning (LoRA, SFT), constrained decoding, vLLM inference, agentic workflows

NLP / Retrieval: RAG, semantic search, NER, Text-to-SQL, large-scale information retrieval

Production ML: Airflow, Docker, AWS ECS, embedding drift detection, LLM output monitoring, ONNX

Team: 12-member team, full hiring lifecycle, campus recruiting (IITM, RV College), mentored 7 promotions

SKILLS

Python • PyTorch • Transformers • vLLM • ONNX • LLM fine-tuning • RAG • NLP • NER • Text-to-SQL • Apache Spark • Airflow • Docker • AWS ECS • Vector Databases • Information Retrieval • Deep Learning

WORK EXPERIENCE

Draup | Bengaluru, India

Oct 2018 – Present (7 yrs 7 mos)

4 promotions: *Data Scientist* > *Senior Data Scientist* > *Data Scientist IV* > *Architect, Data Science* > *Principal Data Scientist*

Principal Data Scientist

Jan 2026 – Present

- Setting technical direction for LLM/AI initiatives across the team. Final reviewer on all major DS decisions before production.

Architect, Data Science

Jan 2023 – Dec 2025

- Grew the team from 6 to 12. Built the hiring process from scratch: wrote the tests, designed the interview rounds, made final calls. Recruited from **IIT Madras, RV College**, and other campuses. Promoted 7 people from intern level up to Senior DS.
- Aligned directly with CPO, CTO, and CEO on ML priorities. The NLP services built by the team **now power several other internal product teams** at Draup.
- Built **CURIE**, our enterprise AI assistant. Combines RAG (30M+ vector embeddings) with Text-to-SQL over 2M+ structured records. Currently used by **260 enterprise clients**. Query accuracy went from 70% to 93% once we got schema structure and business terminology right.
- Rolled out semantic search and NL querying. Feature adoption went from ~30% to ~90% of users, which also cut out **11 manual review roles**. Switched inference to ONNX, ~25% faster on CPU.
- Production monitoring for our models: Mahalanobis distance on mean embeddings to detect data drift, cosine similarity sampling against training set, LLM output/context length distributions tracked over time, and weekly category distribution checks.
- ML pipelines on Airflow + Spark + 3 parallel GPUs. Throughput is roughly **3x what it was** before we moved to this setup.

Data Scientist IV

Jan 2022 – Dec 2022

- Built the News Engine: pre-trained a RoBERTa model on news-domain data and wrote an in-house clustering algorithm to detect emerging global trends in real time.

Senior Data Scientist

Jan 2020 – Dec 2021

- Wrote a dynamic clustering algorithm that determines the number of clusters automatically. Built the Job Responsibility Summarisation system (Seq2Seq BART + custom embedding model) running on 800M+ records.

Data Scientist

Oct 2018 – Dec 2019

- Built an ML classifier (XGBoost + Random Forest) for domain-specific URL categorization, reducing human review effort by ~70%. Shipped **html_tag_annotator** as an open-source Chrome extension to speed up training data labeling (14 GitHub stars).

Calient Technologies | Bengaluru, India

Oct 2017 – Sep 2018

Software Engineer

- Built ML models to classify network device error logs (word2vec + logistic regression). Also built a real-time Java alarm system on top of it.

Cogoport | Mumbai, India

Feb 2017 – Sep 2017

Software Developer

- Designed and built the Bid Request System: data analytics pipeline + REST API in Ruby on Rails.

Techniche e-commerce solutions | India

Apr 2015 – Feb 2017

Software Developer

- Built production e-commerce platforms on Node.js / MEAN stack: user auth, shopping cart, order management.

Hewlett Packard | India

Aug 2013 – Dec 2014

Technical Consultant

- Java development for internal banking systems, focusing on data-heavy SQL pipelines and backend services.

SELECTED PROJECTS

CURIE (Enterprise RAG + Text-to-SQL)

16 months

- Multi-source RAG with 30M+ embeddings + Text-to-SQL over 2M+ structured records. Team of 4. Serves 260 enterprise clients; accuracy 70% to 93%.
- LLM reasoning pipeline that combines vector retrieval and SQL results to answer questions across both unstructured and structured enterprise data.

Job Family Classification (LLM Fine-Tuning + Constrained Decoding)

8 months

- Qwen 0.5B fine-tuned with SFT, served via vLLM. Built a Trie-based constrained decoding layer to guarantee 100% taxonomy adherence. Led a sub-team of 3.
- High-throughput batch pipeline processing hundreds of millions of profiles with Flash Attention for GPU efficiency.

Salary Extraction and Normalization (LLM + NER Pipeline)

6 months

- Pipeline to process 400M+ job descriptions and extract structured compensation data. RoBERTa NER model for entity detection + Gemma-270M fine-tuned for normalization to JSON with currency, ranges, ISO codes.
- Write-up: <https://draup.com/talent/blogs/from-messy-salary-text-to-global-compensation-intelligence>

Job Responsibility Summarisation

4 months

- Seq2Seq BART model + custom embedding model for profession mapping across 800M+ records. Replaced manual annotation and rule-based processing.

WRITING AND OPEN SOURCE

Technical writing:

medium.com/@sachinkalsi | sachinkalsi.github.io/blog

YouTube: youtube.com/@sachin_kalsi — technical deep-dives on Flash Attention, how GPUs work, and core ML concepts

Article: [Constrained Decoding: Forcing LLMs to Respect Your Taxonomy](#)

Open source:

[kmedoids](#) — K-Medoids (PAM) clustering in Python. 24 stars, 22 forks.

[LLM-NER-Offset-Generator](#) — generates NER training data with accurate character offsets using LLMs.

EDUCATION

B.Tech, Computer Science | Visveswaraiah Technological University (VTU)

2013

ADDITIONAL

Languages: English